IAC

Monday, August 17, 2026 10:10 AM

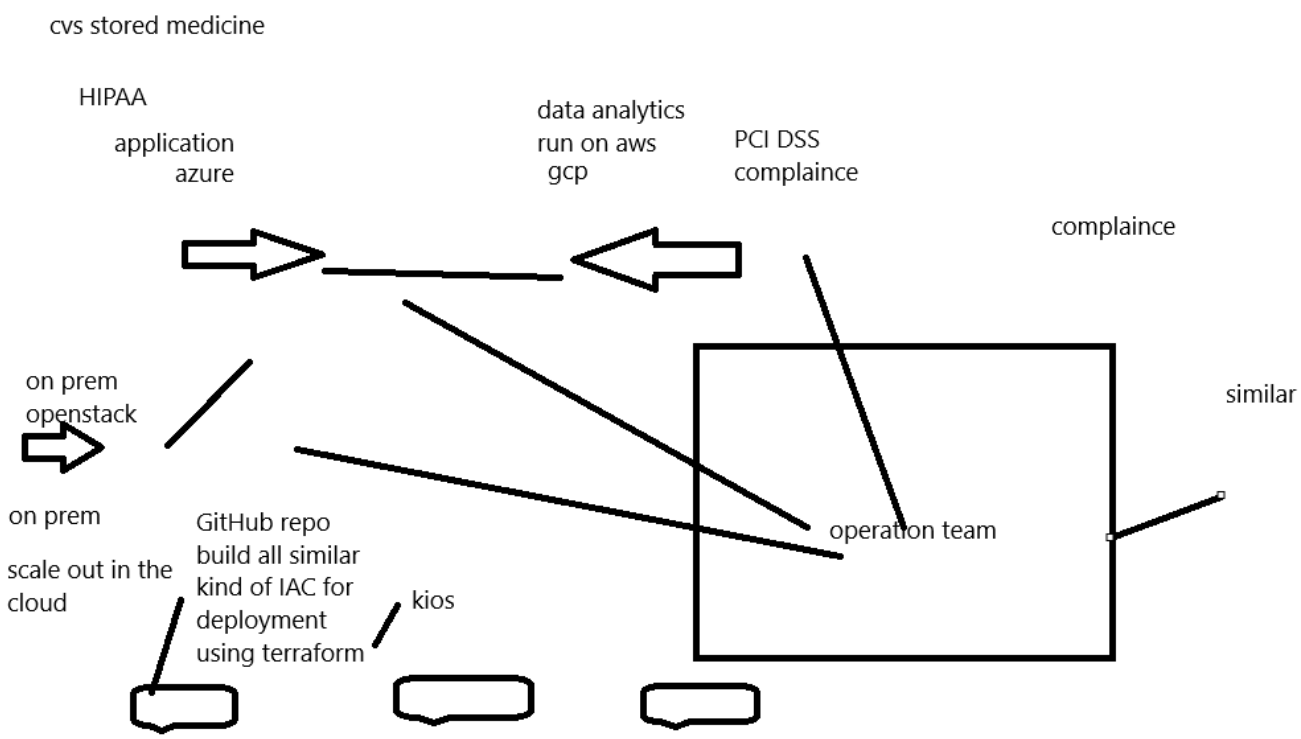
What is the need of infra structure as code.

1. Uniformity across the environment.
2. Faster deployment.
3. Source of truth. Template. Approved
4. Once connected deployed multiple times. Saving the credential in your system. OIDC.

How to avoid security risk in the cloud. If I have an well secured terraform template which will create the resources in the cloud.

We move to an stage where the whole application development lifecycle is managed by SDLC is managed by automation.
Documentation will be generated by system not manual.

DEVOPS
This tools generate those documentation.
Compatibility issue in the cloud.
Organization move toward multi cloud.
1`. Cost
2. Scalability
3. High avaibility



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Sentinel policy

Monday, August 17, 2026 10:37 AM

Policy as code
A developer launching a vm in the cloud
I need to restrict him that he can launch 2 core processor with 4 GB standard d3 v instance
Terraform clou
But with the organization there are certain challenges

1. Client. Xyz
2. Blue print or dry run.

Terraform will always generate an blue print what resource it is creating. Approval from the client.

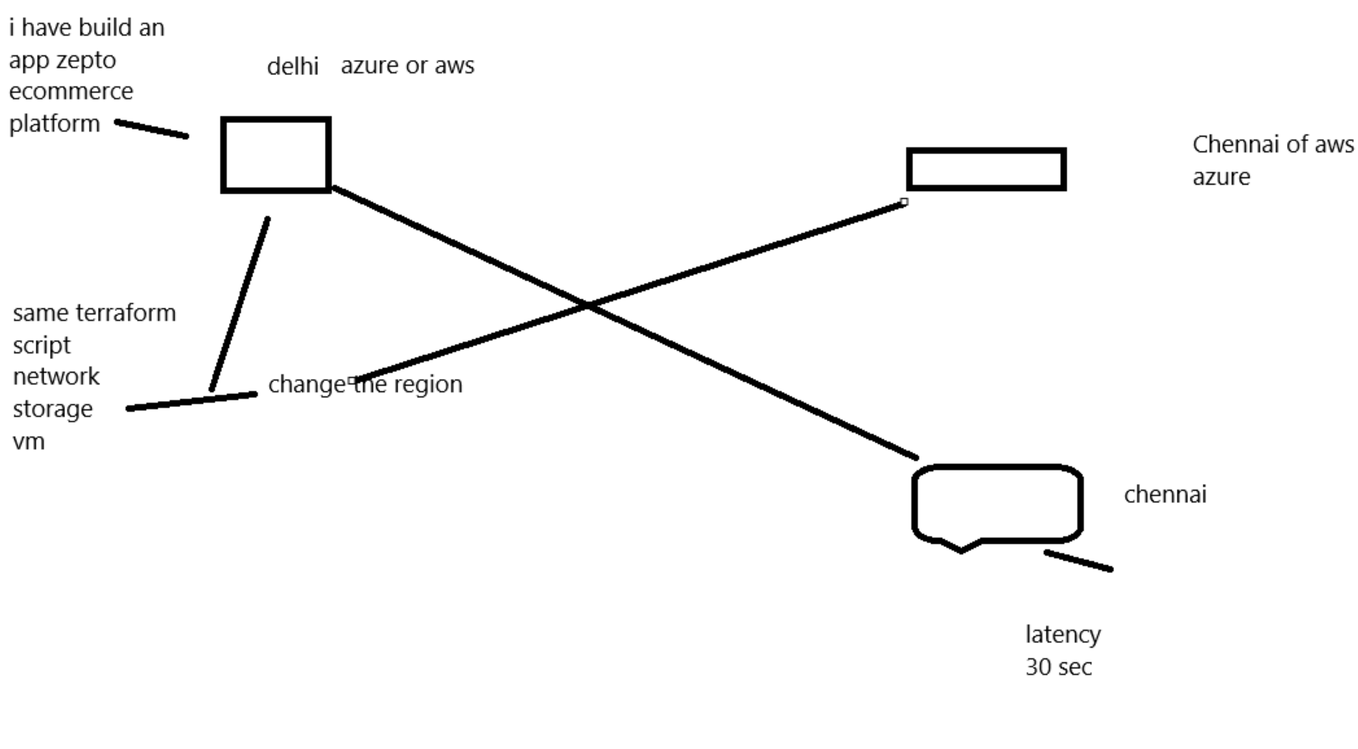
Using terraform we create a virtual machine 2 core processor 8 Gb ram and 250 GB HDD
Azure console
Change it to 4 core 16GB. From the cloud console

But the resource has been created using terraform
Terraform apply
Difference apply it back to the older configuration.
This is called as drift in terraform
If I am managing a rezsources using terraform no changes should happen outside terraform stability
Terraform apply it show the differnce plan

Tags: in the cloud. Identification. Budget
Finance team who put some tags on the production resource. Terraform apply it will go away

Ignore changes "tags" from the console they can add tags.
Ignore changes "vm"
Terraform code is there which will create the network "change the vm configuration"
Ios : glass arm32 processor arm64 bit intel amd quyalcom
I ll not allow any network parameter vm configuration storage standard ssd or premium or ultimate

Scalability



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How to find out an application is well secured

In your nsg you have open port 22 and 3389 0.0.0.0/0
When we write it in code there should be a tools which can provide this valunaribility

Terrafom code to deploy your windows machine.
Password: gitleaks
Code quality tflint

Lab setup

Monday, August 17, 2026 11:34 AM

1. We will create an windows machine in azure and do RDP
2. Inside the windows machine we will install choco package manager. Linux yum apt
3. Terraform
4. Git
5. Authenticate with azure using command line. Aws we have access key and sercret access key.
6. We will create a service principal
7. Using terraform we want to create resource in aws or azure or gcp cloud ibm cloud. We need a tool called as aws cli or azure or gcp cli
8. Terraform git to manage the and cloud service provide cli
   Using az login
   Using service principal
   It will help us to connect any cicd tool azure devops github action or bitbucket pipeline gitlab pipeline. Terraform cloud
   It will be always command line authenticator

   As of now we have the workstation ready with all the authentication parameter

Terraform workflow

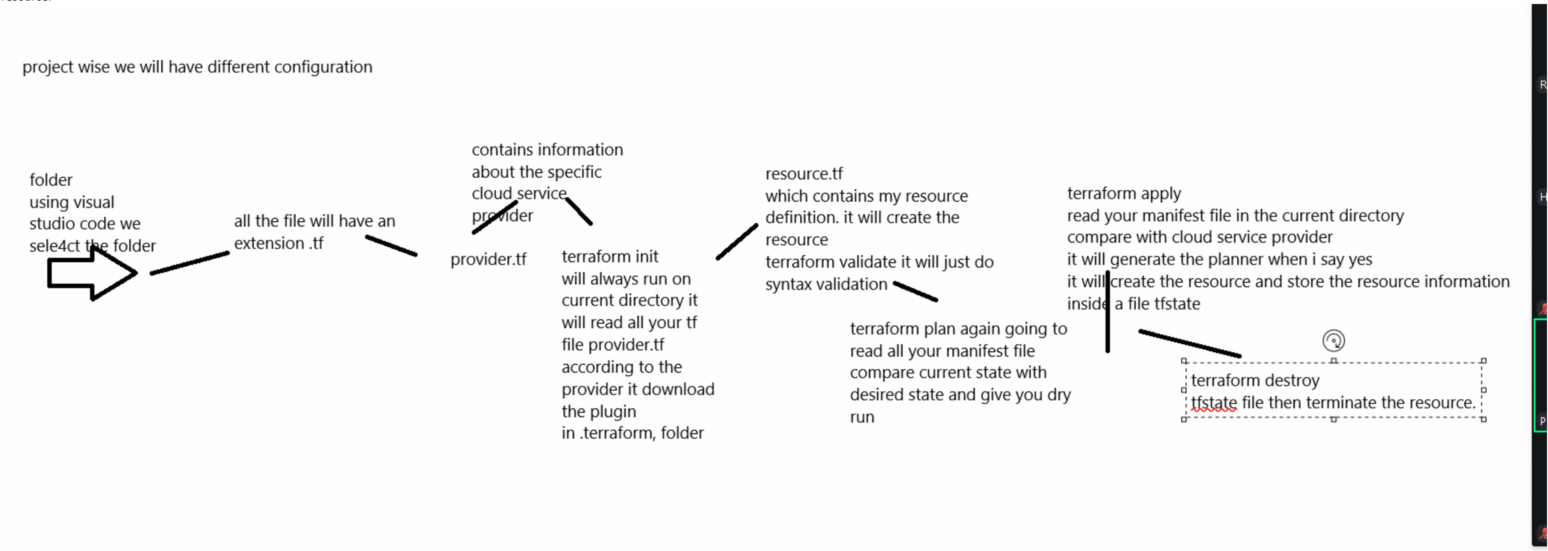
Monday, August 17, 2026 12:49 PM

1. Using terraform what we are going to do? Create infra in the cloud. Which cloud,. Azure
2. Every cloud service IAC code langauge is different. Azure arm template aws use cft cloud formation template gcp use json

Resource group

Monday, August 17, 2026 2:15 PM

We will create a resource in azure.
Everything will be created inside some resource group it is place holder where we group all the resource.



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Slack channel
Or email
Ticket will be updated in jira

Data source block

Monday, August 17, 2026 3:56 PM

As the name suggest calling

Refrence block is used to setup your refrence inside your state file
We have created a resource group inside the rg we need to create a virtual machine.

Recap

Tuesday, August 18, 2026 9:37 AM

We learn about IAC
There are two type of IAC

1. Provisioning tool: terraform ansible: we are going to create resource in the cloud vmware platform entire infra
2. Configuration management tools: ansible chef puppet: we will use to manage the existing resources.

Self checkout machine.
70% of t5rascation world by this self checkout machine.
New feature we have added in the self checkout maching and do an production deployment. Due to some bug stuck up.
Easily do roll back. Ansible to deploy. Update false . Ansible do roll back from all the machine.
75% transaction using self checkout machine. A new feature is develop credit card security has been improved. AES 256 bit . We deploy this feature in self checkout machine. Due to some reason discover. 25% user. You need a system which will deploy your feature in all the machine. Roll back also.
7 days. Wsus

You need to install some predefine software. Encryption you want to encrypt the storage with Customer managed key.
With terraform-ansible works fine

Terraform block: cloud provider plugin
Resource: used for creating resources
Resource nameoftheresource refrenceblock

1. Disctionary in python. Against a key there are multiple values in the named expression format

Refrence block(gopal)
Sad: sadface
Happy: happy face
Vnet : named
Name: nameofthevnet
Location: eastus
Resourcegroup: nameoftheresource
Cidr: 10.0.0.0/16
Key value format
Key it should be always unique

###terraform workflow

1. Terraform init: it will execute inside current directory read your fundamental terraform block. It will download the plugin
2. Terraform validate: this will do syntax validation.
3. Terraform plan: it will read all your manifest file inside the current directory current state compare it with cloud service provide and give you a desired state.
4. Terraform apply: yes it will create the resource in the cloud. It will store the information inside a file tfstate file
5. Terraform destroy: state file destroy all the resources.

Resource group
Mape the vnet with the resource using location and name
Subnet:

1. Vnet and resourcegroup

Graphical

Variables

Tuesday, August 18, 2026 10:22 AM

Reusability of the code is very much important.
Varaibles.
Sekelton of my terraform code. If I can pass varaibles. We can reuse it.

Rg-Vnet-subnet-nsg-nic-storage-vm-loadbalancer which deployed in east us location. Someone from europe accessing my site and getting latency.

Rg-Vnet-subnet-nsg-nic-storage-vm-loadbalance-provide the region. Pass the region name or vm or storage we will use varaibles in terraform.

$ symbol
Var.nameofthevaraible

Varaibles:
Public private

Varaible : scope of varaibles public scope

Where you need to create resource in the cloud . Baja finance.
There are different team who work on azure cloud. Loan
They have created a restriction

Var: scope if global I can use the varaibles outside the folder
Local: the varaible will be used inside the workspace project

Named expression

Tuesday, August 18, 2026 11:35 AM

What is named expression and why you need it

Gopal
Facial: smiles
Body: hand movement
Creative: art

Call only gopal

Nsg

Tuesday, August 18, 2026 12:26 PM

We will create network security group
Security group in aws.

Loops

Tuesday, August 18, 2026 2:15 PM

Every resource block can create single resource vnet subnet or instance
##but there are secanario where we need to create multiple instance with similar configuration.
HA
To do the same we have something called as loop

```
for (int i = 1; i <= 5; i++) {
   System.out.println("Number: " + i);
}
```

Intilization condtion update

Display something value keep on going to run till we fullfill the condtion.

Terraform is not an programming langauge
For each which is loop but it will called as meta argument
Each.key each.value
This going to loop around this key and value
Intilziation I key value 1

String
String will accept single value
If it is production standard d3 instance
Dev standard d4
########for_each

For _each

Tuesday, August 18, 2026 2:50 PM

Every resource block can create a single resource
I want to open port 22 80 8080 443
Same resource block I need to write it 4 times. Duplicating the code.

Dev.tfvars
110=22

Vm

Tuesday, August 18, 2026 3:18 PM

Publicip
Nic card -subnet -privateip

1. Run an shell script: boot straping startup script. I will create a shell which will install apache weserver
2. Paste the public ip in the browser it should show me the test page
3. I want to login inside the instance

Run shell script
Windows I will run powersheel
Only when your system get started for the first time

Sku
Basic: it will create the ip in one single az inside your region
Standard: the public ip will be spread across az data center with in the region
Premium: the public ip will be spread in multiple region

To login to linux machine we can use password based authentication
Key pair based authentication
Sshkeygen in your local system it will generate an private and public key

loops

Tuesday, August 18, 2026 5:42 PM

Recap

Wednesday, August 19, 2026 9:43 AM

Input variable this start with variables
We define what is the type of variables string map of string object type
Inside that we define default value input

We learn about local block
Inside the local block we have put the named expression.

Against a varaible we are defining
Key value pair
Multiple times

For each loop

To call the value in foreach loop key or value
Each.key and each.value

We learn a meta argument called as path.module
It will look the for the file in the current directory

Winvm

Wednesday, August 19, 2026 10:02 AM

I have deployed the vm in east us. All the terraform template configuration is created by taking care of security and organization policy.
Os version are keep on changing . End of the life.
Go to the same template change the os image to an new version
Terraform apply.
I don't need to write the code from the scratch it will increase reusability of the code.
Logistics
Dispatch
Sales: as software upgrade come they are updating the software. And if they run in the cloud
If we have the latest version of the os
They deploy application version1 then they upgrade to version2 they want to deploy the application. It is always a better idea to give them fresh environment
Terraform destroy
Terraform apply which contains a recent version of the os

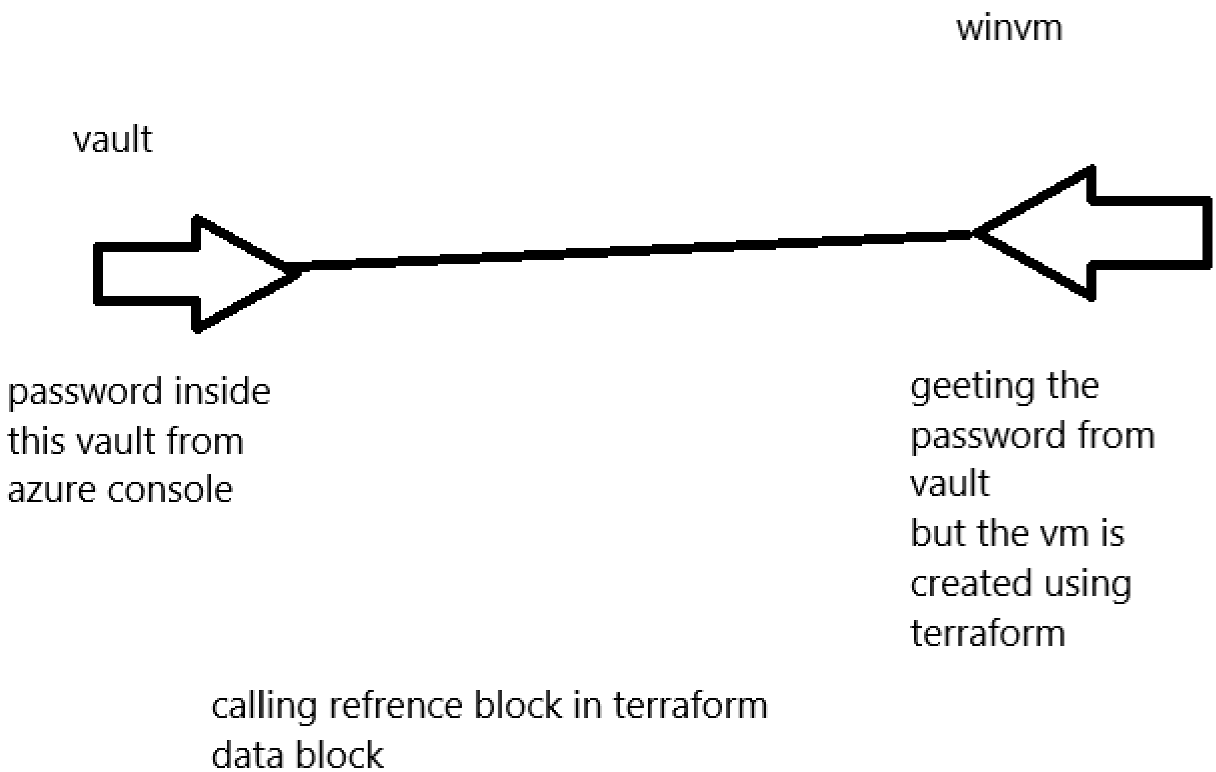
Terraform destroy will destroy your resource in the cloud.
It is not deleting your existing code.
Pravious
Version control system

Take the backup of your systemm
Terraform apply if I change the network parameter like I change the vnet configuration or I chang the port number
It will destroy the vm and storage
Lifecycle {
Prevent destroy}
Resour attach virtual machine data disk
Always keep the data in d:

Vault

Wednesday, August 19, 2026 11:56 AM

1. We will create a vault in azure. Inside that we will store the password of my windows machine
2. But we are creating the vault manually from azure console outside terraform.



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The purpose of the data block in terraform anything you have created outside terraform and I want to call those resource inside your terraform

Permission

Wednesday, August 19, 2026 12:55 PM

We are running terraform script using service principal
Created from the console
Vault we have created using user login
I have to give permission RBAC to the sp so that it can operate the vault

Map variables in terraform

Wednesday, August 19, 2026 3:20 PM

Like an dictionary
In my varaibles files
Default value it can accept single value
When we want to pass on multiple values in key value format.

If it is prod standardd2sv3: 2 core processor and 8 gb ram
Testing 1 core processor 1 gb of ram
Developer 2 core with 4 gb of ram

Loadbalancer
Behind the

Day3 explanation

Thursday, August 20, 2026 9:34 AM

Yes. The important thing to understand is that **the reference block itself does not store information in the state file**. The resource block tells Terraform **what to create and how to identify it**, and after terraform apply, Terraform records the resulting resource information in terraform.tfstate.
Your example:

```
resource "azurerm_resource_group" "myrg" {
name     = "${local.name_prefix}-${var.resource_group_name}"
location = var.location
tags     = local.project_tags
}
```

1. Understand the reference block

This part:

```
resource "azurerm_resource_group" "myrg"
```

has two important names:

```
resource "azurerm_resource_group" "myrg"
|               |
|               └── Your Terraform reference name
|
└── AzureRM resource type
```

azurerm_resource_group

This tells Terraform:

"I want to manage an Azure Resource Group."

myrg

This is a **user-defined Terraform local name/reference**.
You can call it:

```
resource "azurerm_resource_group" "myrg"
```

or:

```
resource "azurerm_resource_group" "gopal_rg"
```

or:

```
resource "azurerm_resource_group" "production"
```

Terraform doesn't care about the name itself, as long as you reference it consistently.

2. What does myrg allow you to do?

Once you define:

```
resource "azurerm_resource_group" "myrg" {
```

you can reference this resource from another Terraform resource.
For example:

```
resource "azurerm_virtual_network" "vnet" {
```

name = "gopal-vnet"
resource_group_name = azurerm_resource_group.myrg.name
location = azurerm_resource_group.myrg.location
address_space = ["10.0.0.0/16"]
}
Here:

azurerm_resource_group.myrg.name
means:
    Get the name attribute from the Resource Group represented by myrg.
And:

azurerm_resource_group.myrg.location
means:
    Get the location attribute from that Resource Group.

3. Your local and variable references

You have:

name = "${local.name_prefix}-${var.resource_group_name}"
Terraform evaluates this expression.
Suppose:

local.name_prefix = "dev-sap"
and:

var.resource_group_name = "gopal-rg"
Terraform creates:

dev-sap-gopal-rg
So:

local.name_prefix
+
var.resource_group_name
↓
dev-sap-gopal-rg
Similarly:

location = var.location
might become:

location = "eastus"
And:

tags = local.project_tags
might become:

Environment = dev
Owner = sap
Project = dispatch

4. What happens when you run terraform plan?
Terraform first reads your configuration.

```
Terraform Code
│
▼
resource "azurerm_resource_group"
│
▼
Evaluate
┌─────────────┼──────────────┐
▼      ▼      ▼
locals   variables   references
│      │      │
└─────────────┼──────────────┘
▼
Desired State
│
▼
terraform plan
```

Terraform determines:

Resource:
azurerm_resource_group.myrg
Name:
dev-sap-gopal-rg
Location:
eastus
Tags:
Environment = dev
Owner = sap
Project = dispatch
Then it compares this desired configuration with the current Terraform state.

5. What happens during terraform apply?
When you execute:

terraform apply
Terraform communicates with Azure:

```
Terraform
│
│ Azure Provider
▼
Microsoft Azure
│
▼
Resource Group Created
```

Azure returns information about the resource.
For example:

Resource ID:
/subscriptions/xxx/resourceGroups/dev-sap-gopal-rg
Name:
dev-sap-gopal-rg
Location:
eastus
Terraform then records the resource's information in its state.

6. What is stored in terraform.tfstate?

Conceptually, Terraform state contains something like:

```
terraform.tfstate
│
└── Resources
│
└── azurerm_resource_group.myrg
│
├── ID
├── Name
├── Location
├── Tags
└── Other attributes
```

The state lets Terraform remember:

> "I created this Azure Resource Group, and this is the real Azure resource associated with azurerm_resource_group.myrg."

The actual state file contains much more information and should **not be edited manually**.

7. Why is the reference important?

Suppose later you create:

```
resource "azurerm_virtual_network" "vnet" {
name             = "gopal-vnet"
resource_group_name = azurerm_resource_group.myrg.name
location          = azurerm_resource_group.myrg.location
address_space      = ["10.0.0.0/16"]
}
```

Terraform sees:

```
azurerm_virtual_network.vnet
│
│ depends on
▼
azurerm_resource_group.myrg
```

So Terraform understands the dependency:

```
Resource Group
│
▼
Virtual Network
```

```
|
▼
Subnet
|
▼
NIC
|
▼
VM
```

This is one of the most important purposes of resource references.

8. The complete workflow

```
┌──────────────────────────────────┐
│ Terraform Configuration  │
│                    │
│ resource "azurerm_..."   │
│ "myrg" {              │
│   name     = ...        │
│   location = ...        │
│   tags     = ...        │
│ }                    │
└────────────────┬─────────────────┘
|
▼
terraform plan
|
▼
┌──────────────────────────────────┐
│    Desired State       │
│                    │
│ dev-sap-gopal-rg        │
│ eastus               │
│ tags                 │
└────────────────┬─────────────────┘
|
▼
terraform apply
|
▼
┌──────────────────────────────────┐
│      Azure           │
│                    │
│ Resource Group Created   │
└────────────────┬─────────────────┘
|
| Resource information
▼
┌──────────────────────────────────┐
```

```
|   terraform.tfstate    |
|                  |
| azurerm_resource_group   |
| .myrg                |
|                  |
| ID                 |
| Name                |
| Location             |
| Tags               |
| Attributes            |
└──────────────────────────────┘
```

The key concept

Think of it this way:

```
resource "azurerm_resource_group" "myrg"
|                 |
|                 |
|                 └── Terraform reference/handle
|
└── What Azure resource Terraform manages
```

Then:

```
azurerm_resource_group.myrg.name
|
▼
Get resource name
```

and:

```
azurerm_resource_group.myrg.location
|
▼
Get resource location
```

After terraform apply:

```
Terraform Resource
|
▼
Azure Resource
|
▼
Terraform State
|
▼
terraform.tfstate
```

So your statement **"using the reference block I am storing all the resource information inside the state file"** is slightly better stated as:

> **The resource block creates a Terraform-managed resource and gives it a reference name. After Terraform creates or updates the resource, Terraform records the resource's identity and attributes in the state file. Other resources can use the reference to establish relationships and dependencies.**

That's the core concept behind Terraform's **resource references + dependency graph + state management**.

Recap

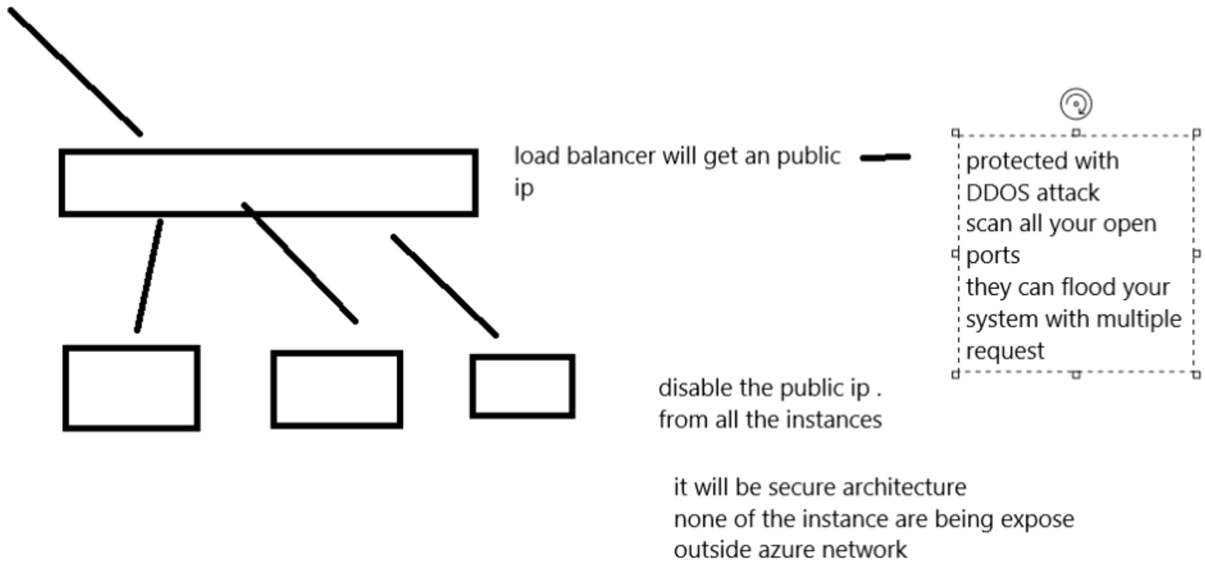
Thursday, August 20, 2026 9:49 AM

Varaibles type called as map varaibles
We have use the same to deploy our reosurce in the cloud

1. We define according to the environment and call the varaibles
2. Var.nameofthevaraible[key]
3. Using the map varaible with for_each loop we have created multiple instance.
4. If I add an default value in the map varaible and do terraform apply it will create similar kind of resources.

loadbalancer

Thursday, August 20, 2026 9:56 AM



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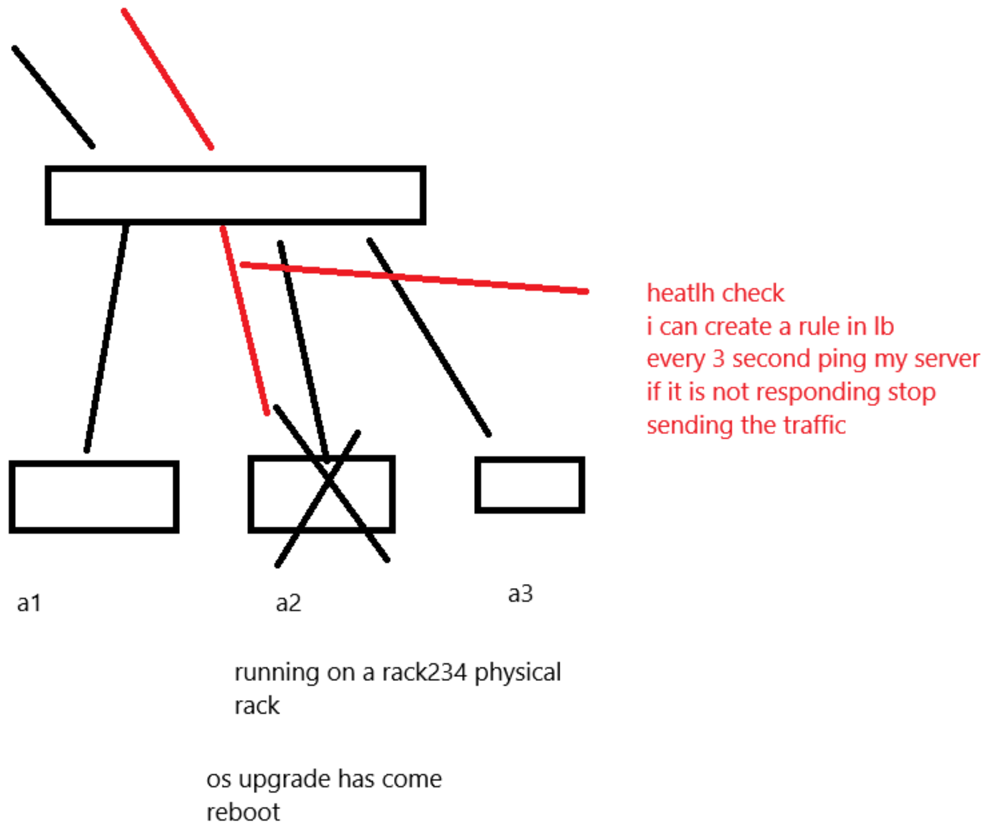
In terraform single line comment #
Multiple line /* */

Lb creation

Thursday, August 20, 2026 10:26 AM

I want to provide access to my appplication server outside the azure network
They do not have any public ip so I cannot access the same. Outside the network
I can create an lb and attach all the network card to the lb

1. Distribute the traffic
2. Natting
3. Port



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1. How to create an loadbalancer in azure.

Lb creating

Thursday, August 20, 2026 10:36 AM

If you know flow of the cloud service same resource block you need to find it from terraform registry

1. Create a public ip.
2. Create the lb and attach the public ip with lb
3. We create backend pool target group
4. Helath checkup
5. Inbound rule
6. Attach those nic card to backendpool where we are grouping all our instances so that lb can send the traffic to that pool
7. Publicip-lb-pool will be attach with your lb

Remote state

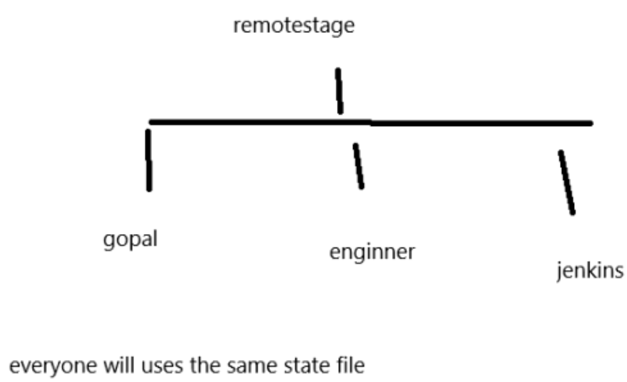
Thursday, August 20, 2026 12:17 PM

Terraform remote state
Currently you terraform apply it will create a tfstate file and stored the log inside the state file
Terraform destroy it will move the state file to tfstate.backup
#if I do the same activity multiple times
On day1 whatever I have done it will be lost
We are storing the state fine inside an local machine and it has read write feature.

I want an storage where if I upload the same file again and again it should not overwrite
Object storage
Aws s3 azure blob storage . If I enable versioing it will allow me to upload the same file multiple times.
Filename:timestamp
Why we need to use remote state
Suppose you and another DevoPs enginner manage the same azure environment if you do not have remote state

Gopal pc	Terraform.tfstate
Other enginner	Terraform.tfstate

Now terraform has two different view of the infrastructure



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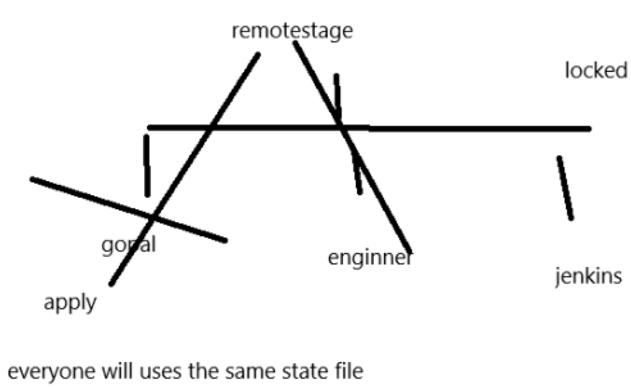

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gopal
terraform apply

locked

harish -terraform apply--wait

anand-terraform apply-wait

apply completed

unlock

this will helps prevent state file corruption
operation

if i have state file in remote and versioning done. audit customer.
i got a request on this date from the email from you PM
terraform apply
this is the state file
team collaboration easily
centralized access to the state file
you lost your machine state will be always there

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Topics

Thursday, August 20, 2026 2:34 PM

Terraform cloud
Modules in terraform

###but before to use terraform cloud ir terraform module you need aware of git version control system.
When you have code you need to manage the same using a version control system.
Git github

Git client tool which you use in your local system to manage your code
Github: it is an remote repo to do colloboration centralized your code.

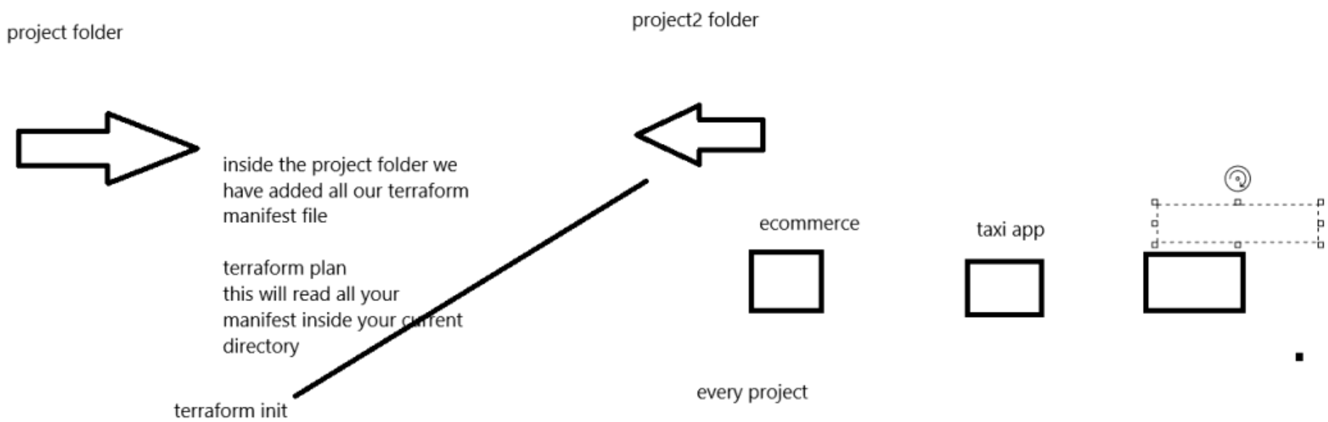
1. Git workflow
2. Git branching

Terraform workflow.

1. Terraform init: terraform will make the current directory as an project
2. Validate: validate code
3. Plan : dry run
4. Apply: apply the code in the cloud
5. Destroy: will read the state file and destroy the resource created in the cloud.

Git workflow

Thursday, August 20, 2026 2:40 PM



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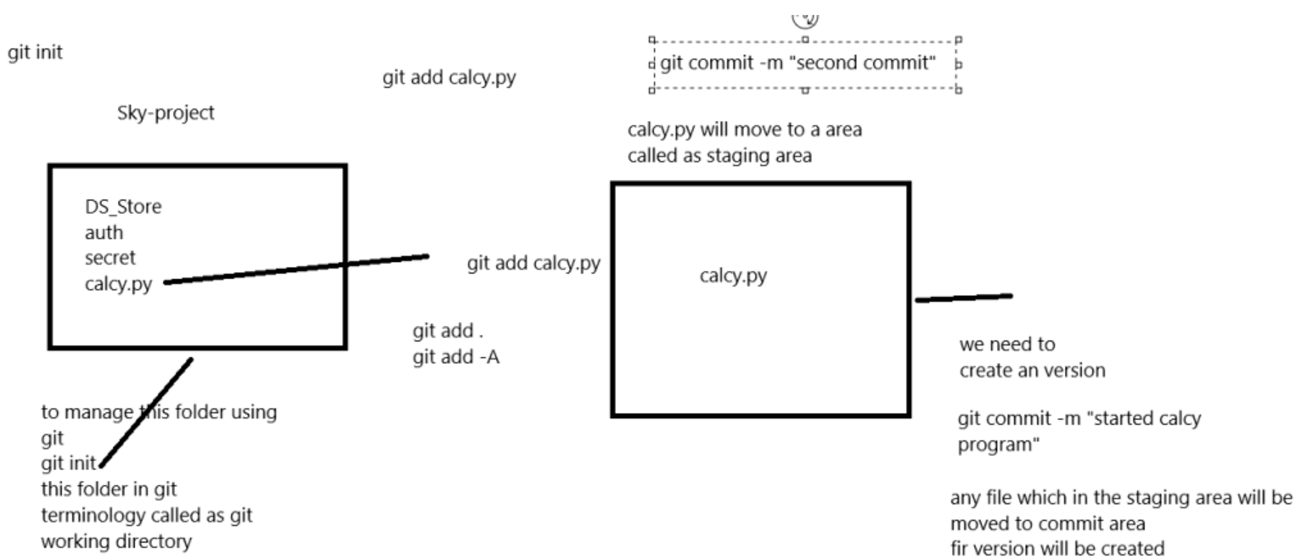


Screen clipping taken: 8/20/2026 3:49 PM

Git client in your cli you cannot see what is the content what commit or compare it
Once we created the version I need to upload the same in github. Then only I can see my changes.
In your laptop you have client used to manage the code.
Git revert

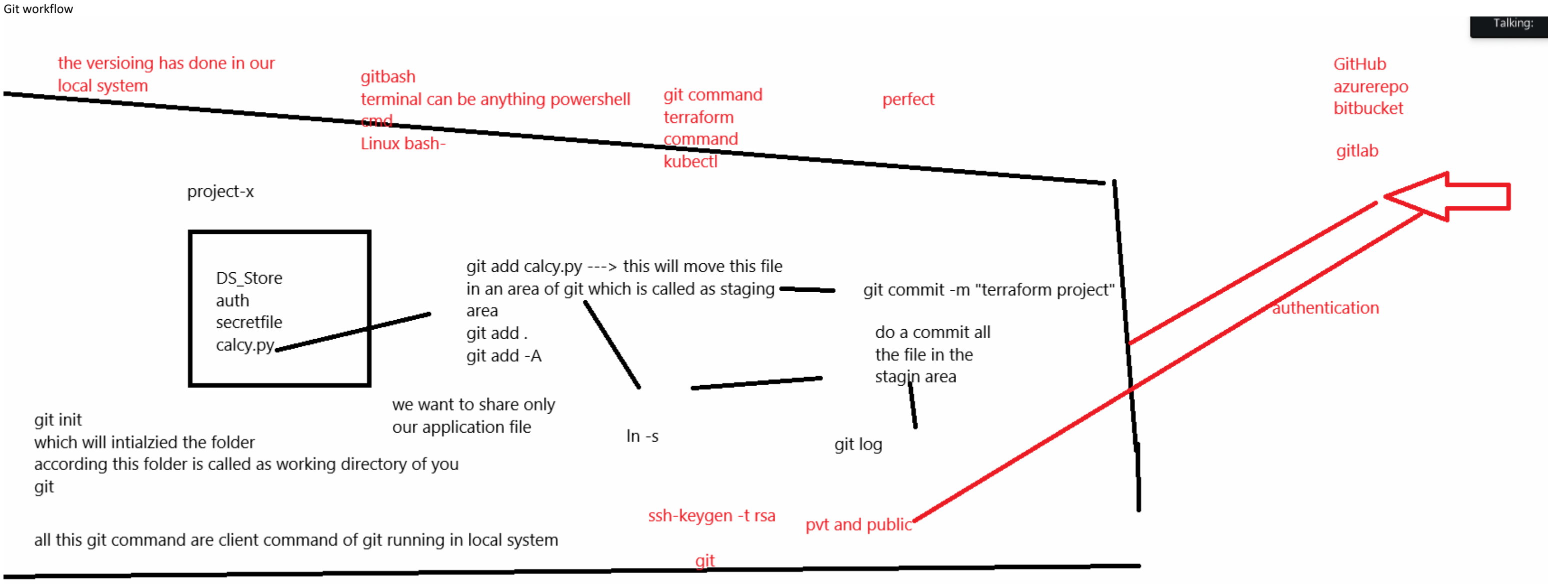
Recap

Friday, August 21, 2026 9:46 AM

We learn about git workflow
In git workflow

1. We learn about global varaibles. User.name and email id.
2. Every commit it will be added using global varaibles.
3. Inside your system if you create n number of application folder and want to manage using git global varaibles will be automatically attached.

Git workflow



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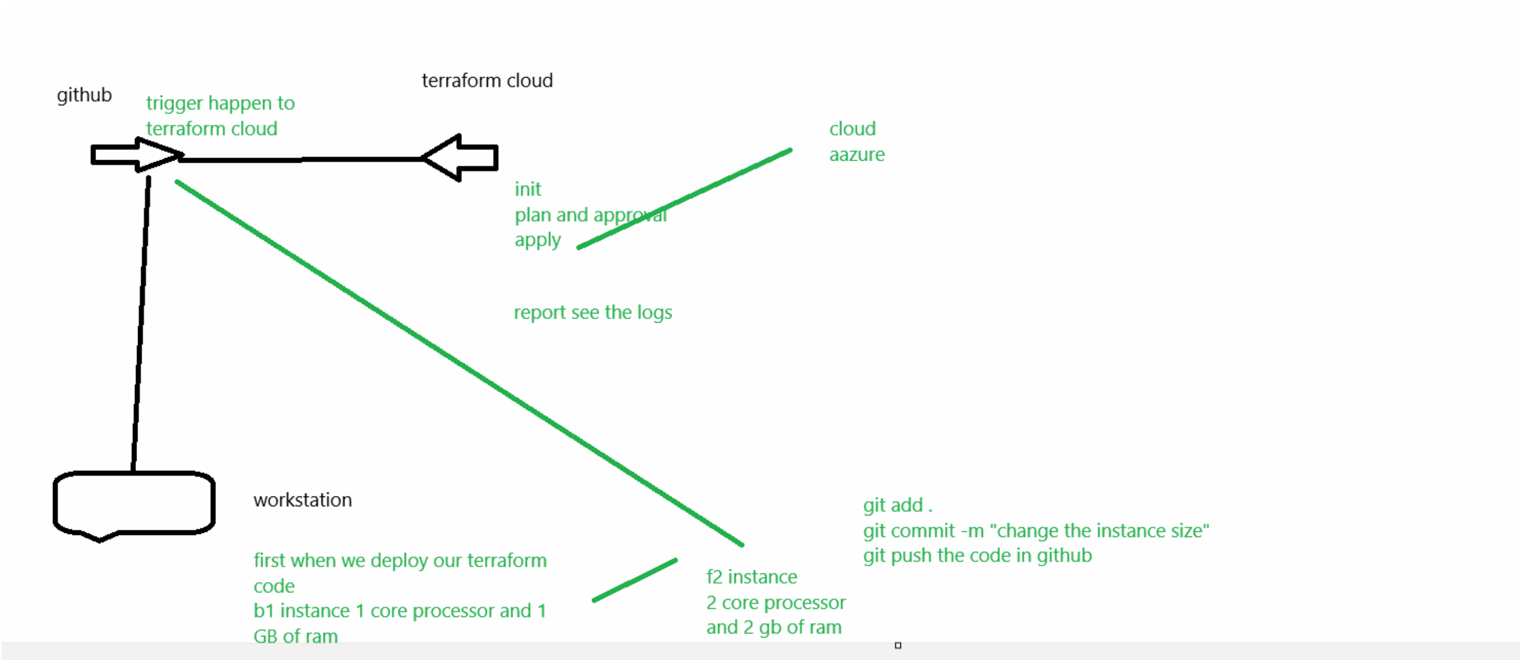
Terraform cloud

Friday, August 21, 2026 10:15 AM



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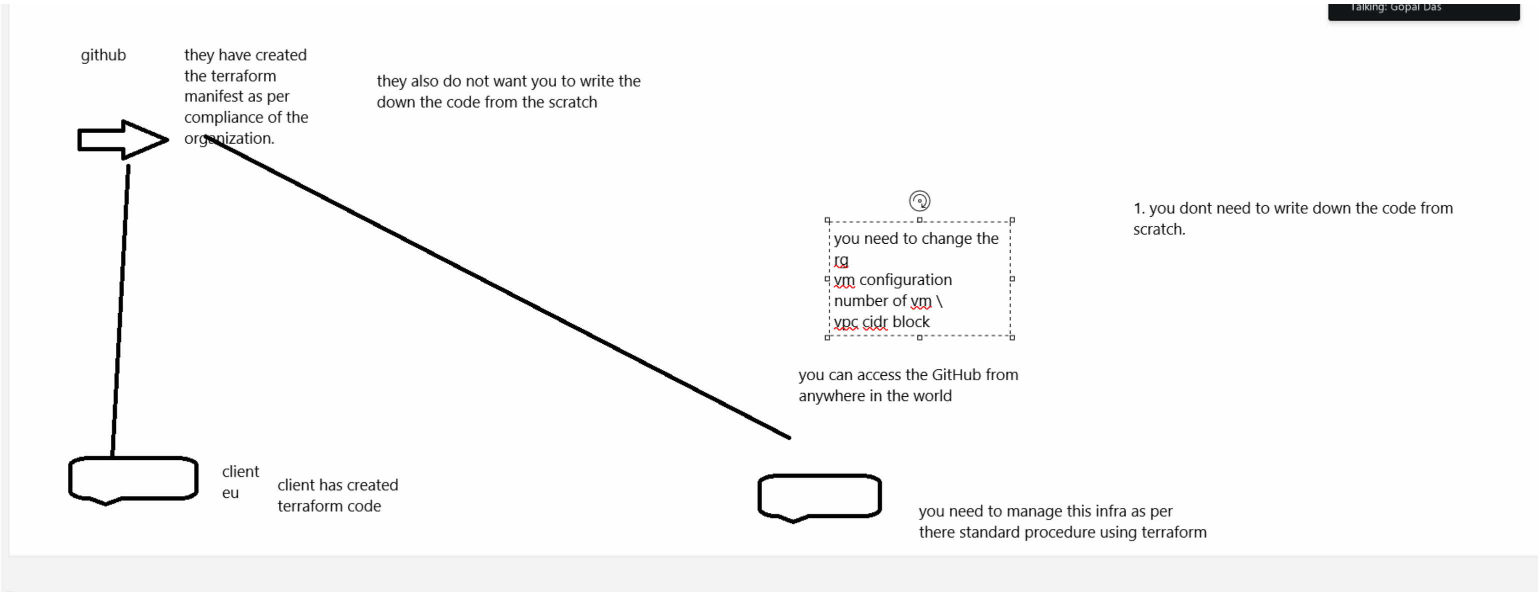
Terraform cloud

Friday, August 21, 2026 11:47 AM

1. We will enable pricing
2. Auto apply

Modules in terraform

Friday, August 21, 2026 2:09 PM



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Modules

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To create networking
1.vnet
2. Subnet
3. Nsg
##########i upload the skeleton of the resource block in a version control. They need to follow this standard

Ops1 devops1